ENCE361: Personal step counter project

Introduction

The goal of this project is to implement an embedded sysytem to record a user's steps and estimate the distance travelled by the user. The step counter will be powered by a STM32C071 nucleo development board, connected to an expansion board. The project will involve developing firmware for the STM32C071 to record and display step data via a graphical user interface. Like professional embedded systems projects, this project will require both technical knowledge and interpersonal skills to complete successfully.

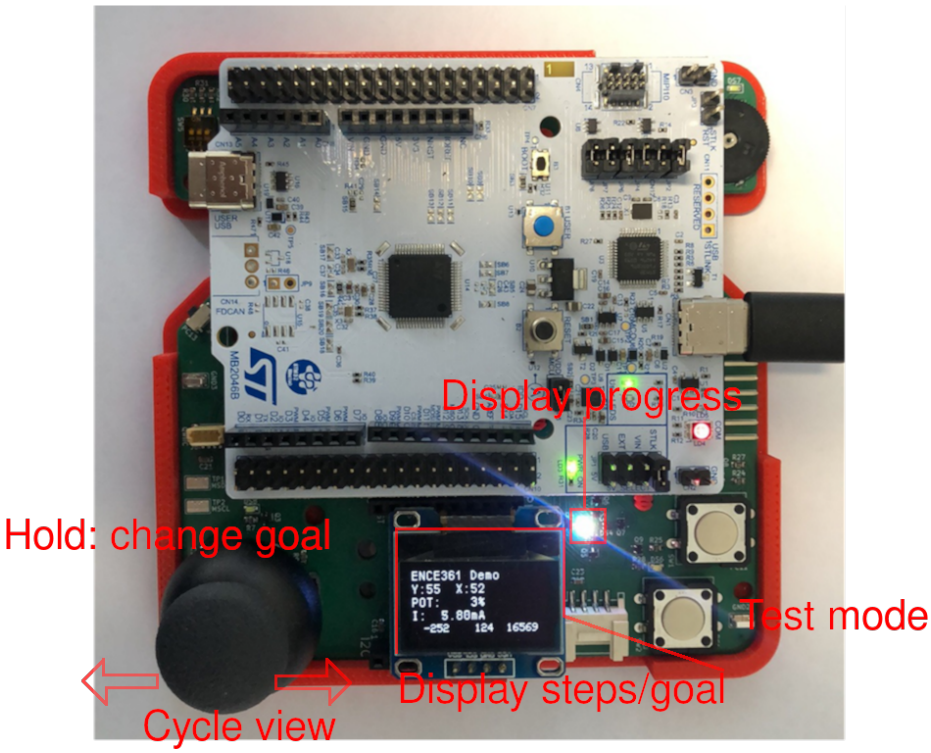

Figure 1: Expansion board for the STM32 nucleo board.

Project organisation

The project is completed in groups of two, and directly builds on the labs from term 1. Lab exercises in term 1 provide the basic foundation of how to program the board and understanding the key components of the system. At the end of term 1 there will be a milestone assessment where you must demonstrate base functionality of sampling analogue and digital inputs, as well as simple serial debugging, followed by a another milestone partway through term 2. A final demonstration of the overall project will be near the end of term 2 where you must show your complete implementation of the system.

There are labsheets in weeks 1-4, 8 to allow a more hand-holding walk through the broad implementation of some aspects of the project. For milestone one in particular you are expected to keep up to date with the first round (weeks 1-4) of lab sheets. If you are not completing the lab sheets within the supported lab times you will need to ensure this is done outside of lab time.

Assessment

The project is worth 40 % of your final grade for ENCE361. Marks for the project will be awarded for the whole group, but will factor in peer-assessed contribution for effort towards the milestones, effort towards the final demonstration, and attitude and expectations to working as a team. Your peer assessment must be submitted by the time you submit your final report for this project via the Learn peer assessment item. To maximise the chance of successful completion, the project is structured in terms of three sequential milestones, as follows:

Component	Due	Weight	Notes
Milestone 1	Week 5	3 %	In-lab, requires code on git
Milestone 2	Week 7	5 %	In-lab, requires code on git
Final demonstration	Week 10	15 %	In-lab, earliest allocated slot
Code submission	6pm after demonstration lab	10 %	Marked from Gitlab repo
Report	Week 12	7 %	One per pair
Total:		40 %	

Marking rubrics

For milestones one and two you will be required to demonstrate the functionality of your firmware in the labs to a TA. Your implementation will be analysed to determine if it meets the requirements laid out in the applicable specification document.

The code submission will be marked for (with equal weighting):

1. Readability (e.g. appropriate names, commenting, abstractions, function breakdown)
2. Modularity and structure (e.g. appropriate split of responsibilities, loose coupling, encapsulation, layering)
3. Scheduling and interrupts (e.g. interrupt-based kernel, sensible task split, task frequencies, use of ISRs, handling shared resources)

Documentation will be required to be in the form of a short report completed by each team due in week 12. You should expect to discuss your solutions and firmware implementation

Resources

- Hardware: Each group will be loaned a STM32 nucleo board, USB A-C cable, and an expansion board. These must be returned at the end of the project. Failure to return the hardware will result in a 5% mark deduction. There will be battery packs available for testing in the labs.
- Source code: Code from labs and lectures are available on Learn, and significant examples are available from STMicroelectronics. You are able to use code from here or any other source provided you:
  1. Explicitly acknowledge any source of code used, typically through commenting within your source files, AND
  2. Take responsibility for any code you use: this responsibility includes changing any commenting and syntax to form a cohesive source for all of your code.
- Technical support: Support from tutors and lab supervisors (Philipp Hof, Randy Hampton and Diego Ramirez) will be available throughout the project. Outside of lab times, support will be available via the ENCE361 Learn forum, where students will be able to ask (and answer) questions related to the course. Professional development, in the form of introductory lab exercises and lecture content, will be provided and will be arranged so that key concepts are covered near the time that they are likely to be needed in the project. Extensive technical documentation is also available from component manufacturers such as STMicroelectronics and Analog Devices, amongst others.
- IDE: STM32CubeMX + STM32CubeIDE will be used for this project. The SDK (files which allow your code to be deployed on the STM32C071) are installed on the lab computers for you. All of these are also available for download onto your own devices from STMicroelectronics, but will require you to register an account. Please note that the versions of CubeMX, CubeIDE, **and the C07 1 firmware on the lab computers may not be the most recent version**, so it is

recommended to manually download the same versions, or working on your own devices may make your software incompatible with the lab computers.

- Further documents will be made available specifying:
  1. Milestone 1
  2. Milestone 2
  3. Full working demonstration requirements
  4. Report guidelines

Group allocation

By now you are all working in groups. It is important that you continue to work effectively in your group, including communicating about how and when you will be working together to achieve the milestones and project as a whole. Up to one third of the report and code mark (i.e. ≈5% of the overall course) will be determined by the peer assessment on Learn.

A git repository has been be created for each group, accessible at `https://eng-git.canterbury.ac.nz`. Code submissions will be automatically taken from your group repository on the day you demonstrate.

This is a new project, we welcome feedback as we go